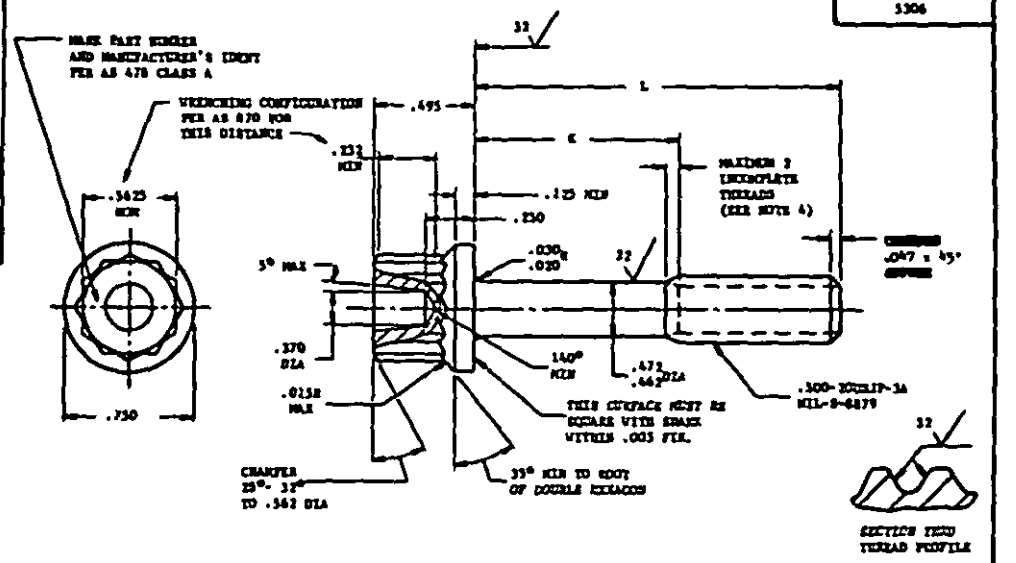
# MS9890 (ASG)

FED. SUP CLASS 5306




| PART NO. | L | K | APPROX. WEIGHT LB/100 |
|---|---|---|---|
| MS9890-06 | .812 | .103- .125 | 7.99 |
| MS9890-07 | .875 | .103- .125 | 8.25 |
| MS9890-08 | .938 | .103- .125 | 8.59 |
| MS9890-07 | 1.000 | .103- .125 | 8.90 |
| MS9890-08 | 1.062 | .103- .125 | 9.22 |
| MS9890-09 | 1.125 | .103- .125 | 9.54 |
| MS9890-10 | 1.188 | .103- .125 | 9.86 |
| MS9890-11 | 1.250 | .103- .125 | 10.18 |
| MS9890-12 | 1.312 | .103- .125 | 10.48 |
| MS9890-13 | 1.375 | .103- .125 | 10.78 |
| MS9890-14 | 1.438 | .125- .188 | 11.09 |
| MS9890-15 | 1.500 | .190- .250 | 11.40 |
| MS9890-16 | 1.562 | .252- .312 | 11.71 |
| MS9890-17 | 1.625 | .313- .373 | 12.02 |
| MS9890-18 | 1.688 | .378- .438 | 12.33 |
| MS9890-19 | 1.750 | .440- .500 | 12.64 |
| MS9890-20 | 1.812 | .502- .562 | 12.93 |
| MS9890-21 | 1.875 | .563- .625 | 13.24 |
| MS9890-22 | 1.938 | .625- .688 | 13.53 |
| MS9890-23 | 2.000 | .690- .750 | 13.86 |
| MS9890-24 | 2.125 | .813- .875 | 14.45 |
| MS9890-25 | 2.250 | .940-1.000 | 15.06 |
| MS9890-26 | 2.375 | 1.065-1.125 | 15.68 |
| MS9890-27 | 2.500 | 1.190-1.250 | 16.30 |
| MS9890-28 | 2.625 | 1.315-1.375 | 16.91 |
| MS9890-29 | 2.750 | 1.440-1.500 | 17.52 |
| MS9890-30 | 2.875 | 1.565-1.625 | 18.13 |
| MS9890-31 | 3.000 | 1.690-1.750 | 18.74 |
| MS9890-32 | 3.125 | 1.815-1.875 | 19.35 |
| MS9890-33 | 3.250 | 1.940-2.000 | 19.96 |
| MS9890-34 | 3.375 | 2.065-2.125 | 20.58 |
| MS9890-35 | 3.500 | 2.190-2.250 | 21.20 |
| MS9890-36 | 3.625 | 2.315-2.375 | 21.81 |
| MS9890-37 | 3.750 | 2.440-2.500 | 22.42 |
| MS9890-38 | 3.875 | 2.565-2.625 | 23.04 |
| MS9890-39 | 4.000 | 2.690-2.750 | 23.66 |
| MS9890-40 | 4.125 | 2.815-2.875 | 24.27 |
| MS9890-41 | 4.250 | 2.940-3.000 | 24.88 |
| MS9890-42 | 4.375 | 3.065-3.125 | 25.49 |
| MS9890-43 | 4.500 | 3.190-3.250 | 26.10 |
| MS9890-44 | 4.625 | 3.315-3.375 | 26.71 |
| MS9890-45 | 4.750 | 3.440-3.500 | 27.32 |
| MS9890-46 | 4.875 | 3.565-3.625 | 27.93 |
| MS9890-47 | 5.000 | 3.690-3.750 | 28.59 |
| MS9890-48 | 5.125 | 3.815-3.875 | 29.20 |
| MS9890-49 | 5.250 | 3.940-4.000 | 29.81 |
| MS9890-50 | 5.375 | 4.065-4.125 | 30.42 |
| MS9890-51 | 5.500 | 4.190-4.250 | 31.03 |
| MS9890-52 | 5.625 | 4.315-4.375 | 31.64 |
| MS9890-53 | 5.750 | 4.440-4.500 | 32.25 |
| MS9890-54 | 5.875 | 4.565-4.625 | 32.86 |
| MS9890-55 | 6.000 | 4.690-4.750 | 33.47 |

1. SHANK SHALL BE STRAIGHT WITHIN .002 PER INCH OF BOLT LENGTH.
2. THE CONCENTRICITY OF THREAD PD IN RELATION TO THE SHANK SHALL BE WITHIN .006 FIR.
3. THE CONCENTRICITY OF THE SHANK IN RELATION TO THE WASHER FACE DIAMETER AND DOUBLE HEXAGON OD SHALL BE WITHIN .012 FIR.
4. INCOMPLETE THREADS MAY TO ENTER FILLET.
5. MATERIAL: CORROSION RESISTANT STEEL AMS 5616.
6. MANUFACTURING SPECIFICATION: AMS 7470 EXCEPT HEAD SHALL BE UPSET.
7. HEAD TO SHANK FILLET SHALL BE COLD ROLLED AFTER HEAT TREATMENT TO REMOVE ALL VISUAL EVIDENCE OF GRINDING OR TOOL MARKS.
8. HARDNESS: ROCKWELL C32-38.
9. SURFACE TEXTURE: USAS B46.1-1962. UNLESS OTHERWISE SPECIFIED, SURFACE TO BE 125 MICROINCHES EXCEPT UPSET HEAD.
10. MAGNETIC PARTICLE INSPECTION PER AMS 2640.
11. BREAK SHARP EDGES .003-.015 UNLESS OTHERWISE SPECIFIED.
12. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: LINEAR DIMENSIONS ± .010, ANGULAR DIMENSIONS ± 5°.
13. DO NOT USE CRASSIFIED PART NUMBERS.

AS & AMS ARE SOCIETY OF AUTOMOTIVE ENGINEERS, INC. PUBLICATIONS.
THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE MILITARY SERVICES BY THE SAE AEROSPACE PART STANDARDS DIVISION.

| P.A. | TITLE | MILITARY STANDARD |
|---|---|---|
| CCAF - 11<br>Other Cust<br>Navy - AS | BOLT, MACHINE - DOUBLE HEXAGON EXTENDED WASHER HEAD, AMS 5616, .500-20 UNJF-3A | MS9890 (ASG) |
| PROCUREMENT SPECIFICATION | SUPERSEDES | SHEET 1 OF 1 |

DD FORM 672-1 (Electronic coordination) ASD use only

Project No. 5306-F010

APPROVED 1 May 69 REVISED

This military standard is approved by the Department of the Air Force and is mandatory for use by them in all Air Force procurement activities. Other military activities are required to employ this standard where suitable.

Review activities: Navy - AS, MC

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